AWS EC2 Direct Root Login via SSH

AWS Academy Sandbox • Ubuntu • MobaXterm • Root Password Authentication

Practical step-by-step documentation

This document explains how to start an AWS Academy Sandbox, launch or access an Ubuntu EC2 instance, connect with MobaXterm, and configure password-based SSH login for the root account in a controlled lab. Each step includes a simple explanation, commands where required, expected checks, and a place to insert a genuine screenshot.

> **Security note:** Direct root login over SSH using a password weakens server security and is **not recommended for production**. Use this only in an isolated, instructor-approved lab. Never expose password-based root SSH to the public internet. At the end, restore safer settings and use the normal Ubuntu account with **sudo**.

Objectives

- Understand how to open an AWS Academy Sandbox and access its temporary AWS environment.
- Use the EC2 console to identify an Ubuntu virtual machine and its public IPv4 address.
- Connect to the instance using MobaXterm and the standard Ubuntu login.
- Set a root account password and understand the SSH configuration needed for a lab demonstration.
- Verify the connection and then restore safer SSH settings.

Requirements and tools

Item	Purpose
AWS Academy Sandbox	Provides temporary access to AWS lab resources.
Ubuntu EC2 instance	Linux virtual machine used for the practical.
MobaXterm	SSH client used from Windows to connect to the instance.
SSH private key	Authenticates the initial connection if the instance was created for key-based access.
Terminal / shell	Runs Linux commands with appropriate privileges.
Internet access and port 22	Required for SSH; restrict the source IP in the EC2 security group.

> **Before you begin:** Use only an instance you are authorized to administer. If your instructor provides a preconfigured instance, follow the course instructions rather than creating extra resources. AWS Academy accounts are temporary and may have restrictions.

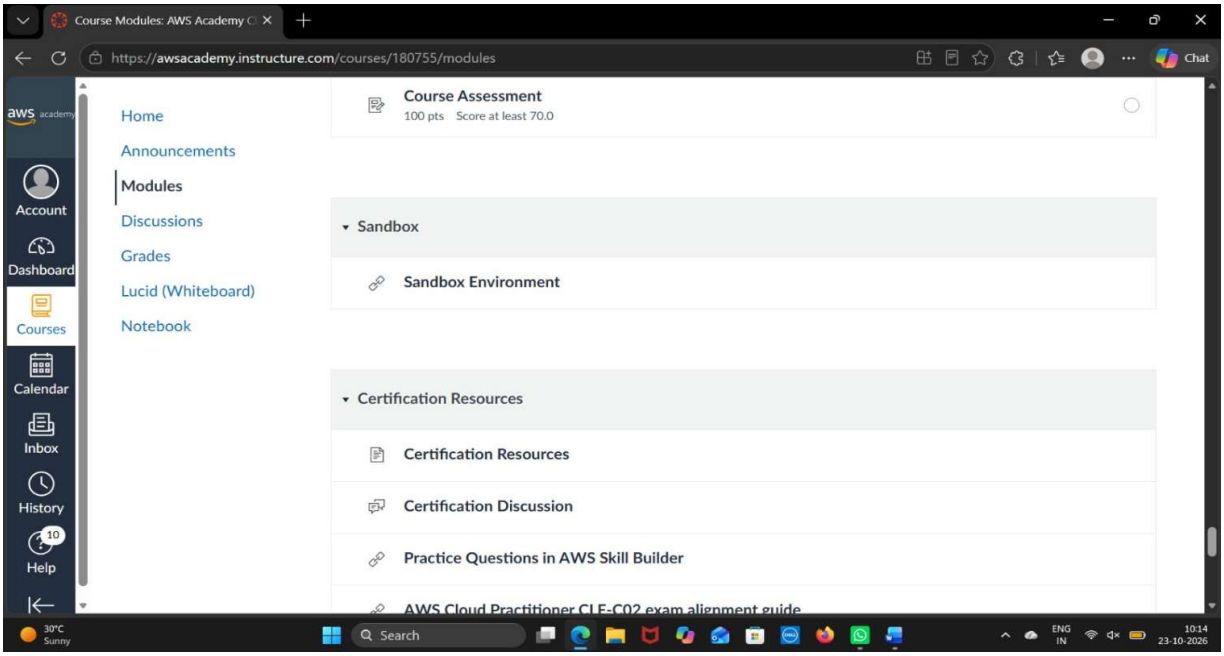
Part 1 — Start the AWS Academy Sandbox

Open the lab environment that provides temporary AWS access.

Step 1 — Open the Sandbox Environment module

In the AWS Academy course, open **Modules** and scroll to the **Sandbox** section. Select **Sandbox Environment** to open the lab launcher. The exact course layout may vary by class.

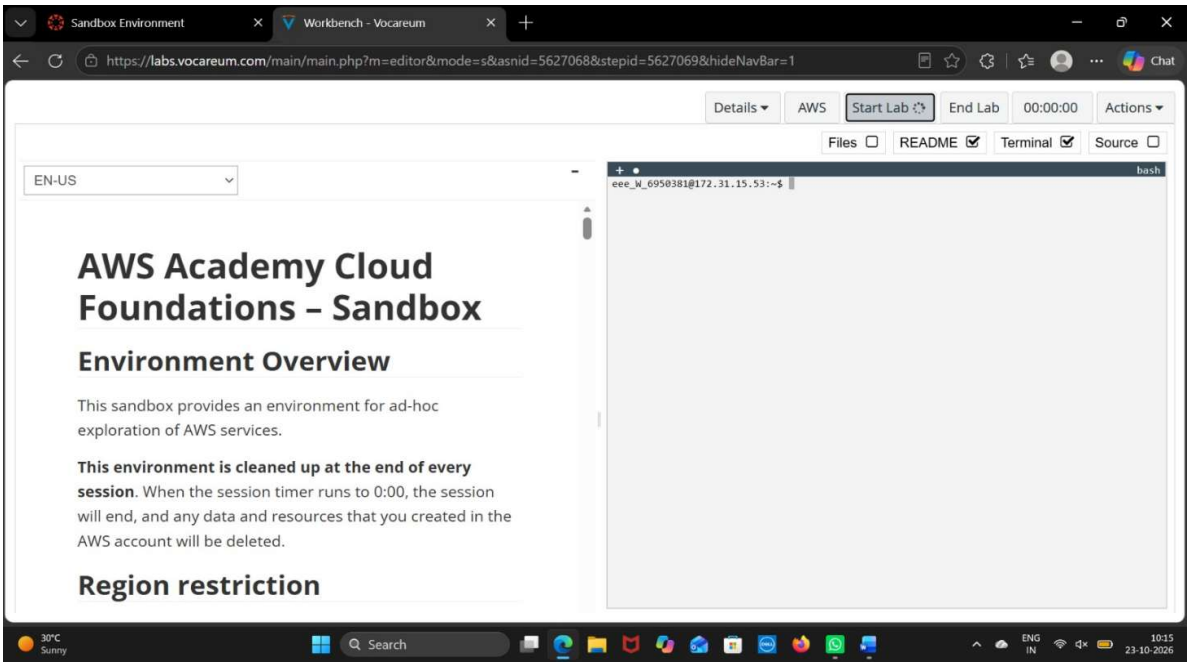
Expected result: The Sandbox Environment page or lab launcher opens.



Step 2 — Start the lab

On the Sandbox page, click the available start or launch control. Wait for the lab status to indicate that the environment is ready. If a timer is shown, note the remaining session time.

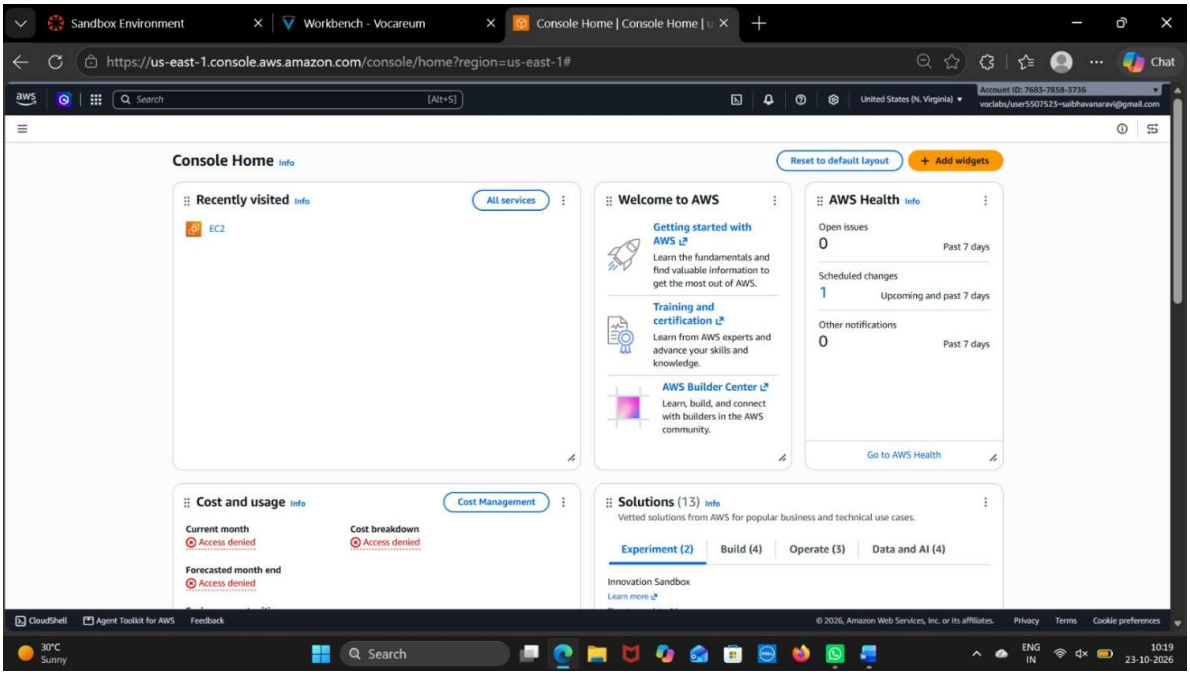
Expected result: The lab status changes to ready/started and the AWS console access becomes available.



Step 3 — Open the AWS console

Use the console link or button provided by the AWS Academy lab. Do not use personal AWS credentials unless the instructor specifically directs you to do so.

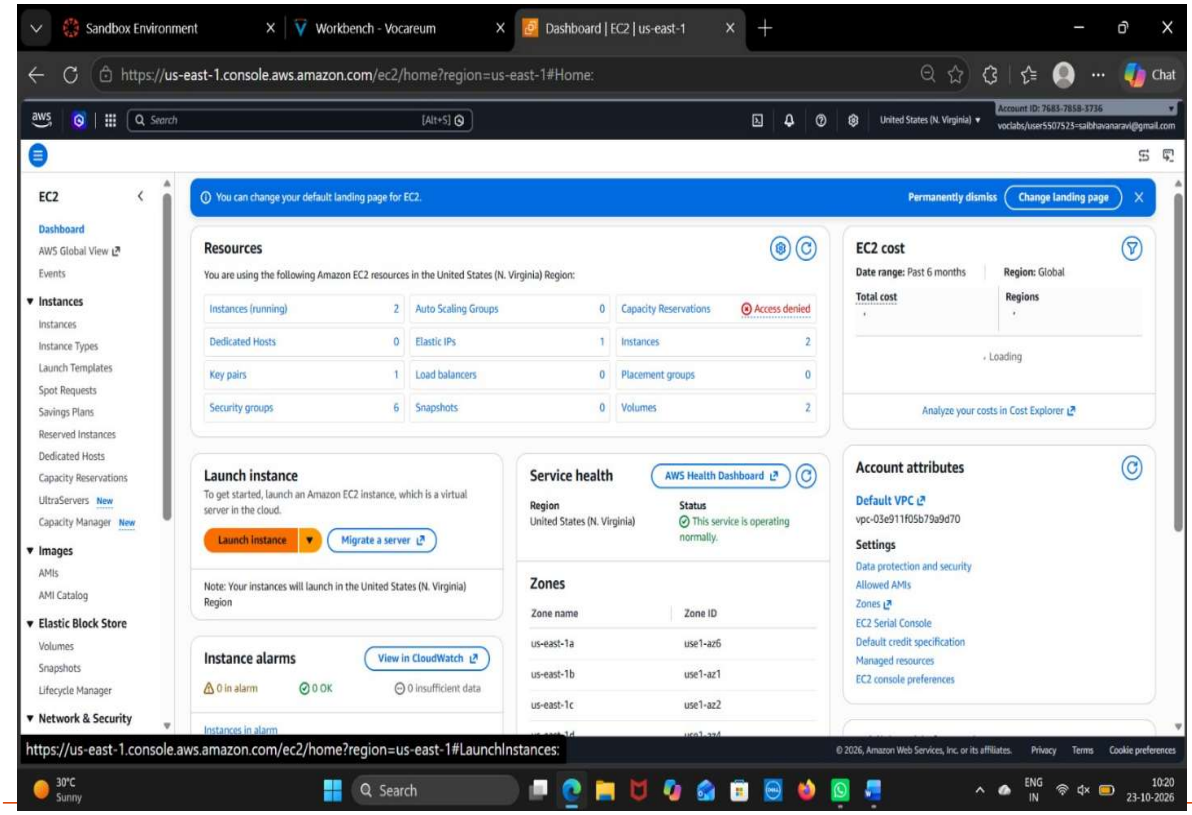
Expected result: The AWS Management Console opens in the lab account.



Step 4 — Review the EC2 dashboard

The EC2 dashboard summarizes the resources in the region. At this point there are no running instances.

Click Launch Instance to begin creating the server.



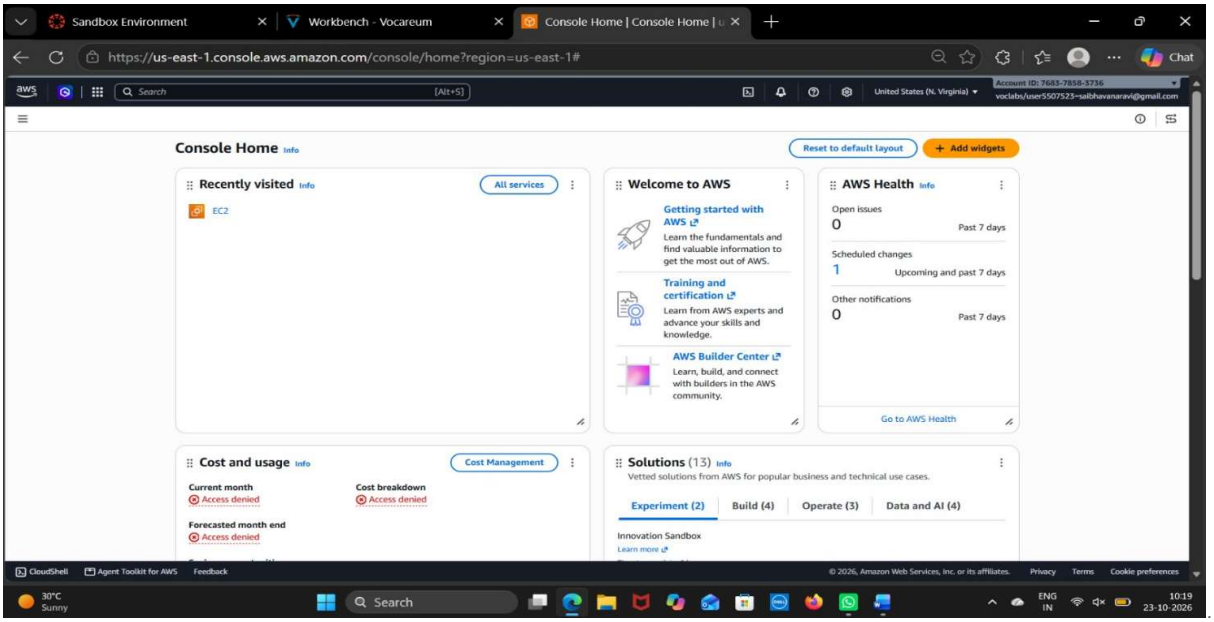
Part 2 — Launch and prepare the Ubuntu EC2 instance

Create or locate the lab virtual machine and check that SSH access is possible.

Step 5 — Open EC2

In the AWS Management Console, search for **EC2** and open the EC2 service. Select **Instances** from the navigation panel.

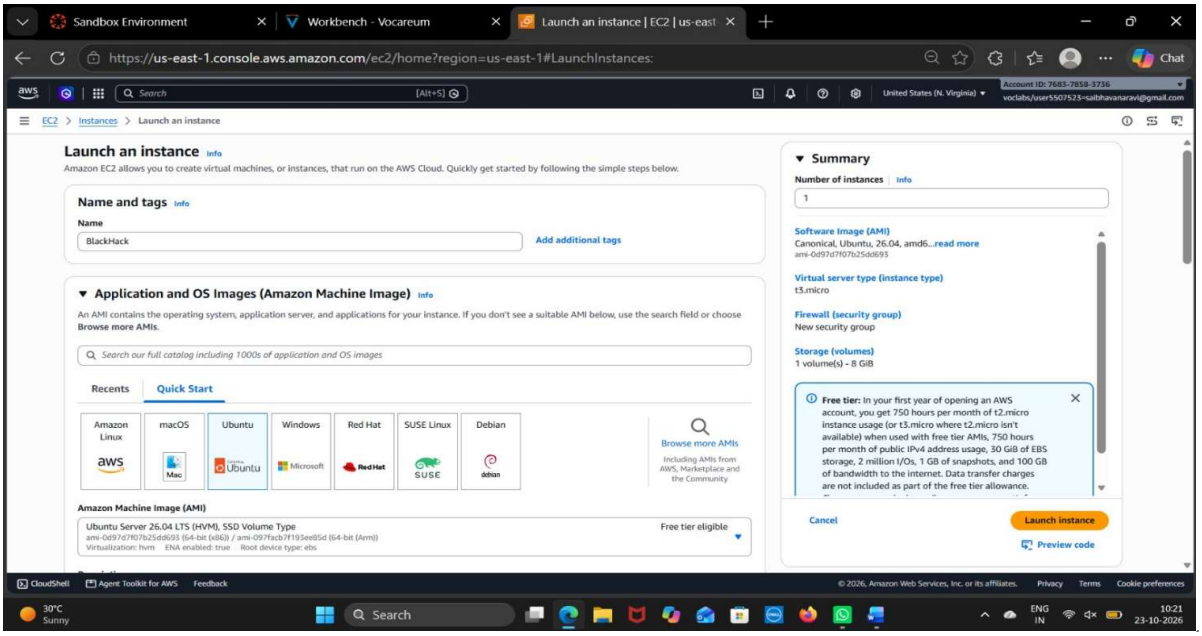
Expected result: The EC2 Instances page lists the lab instances, if any.

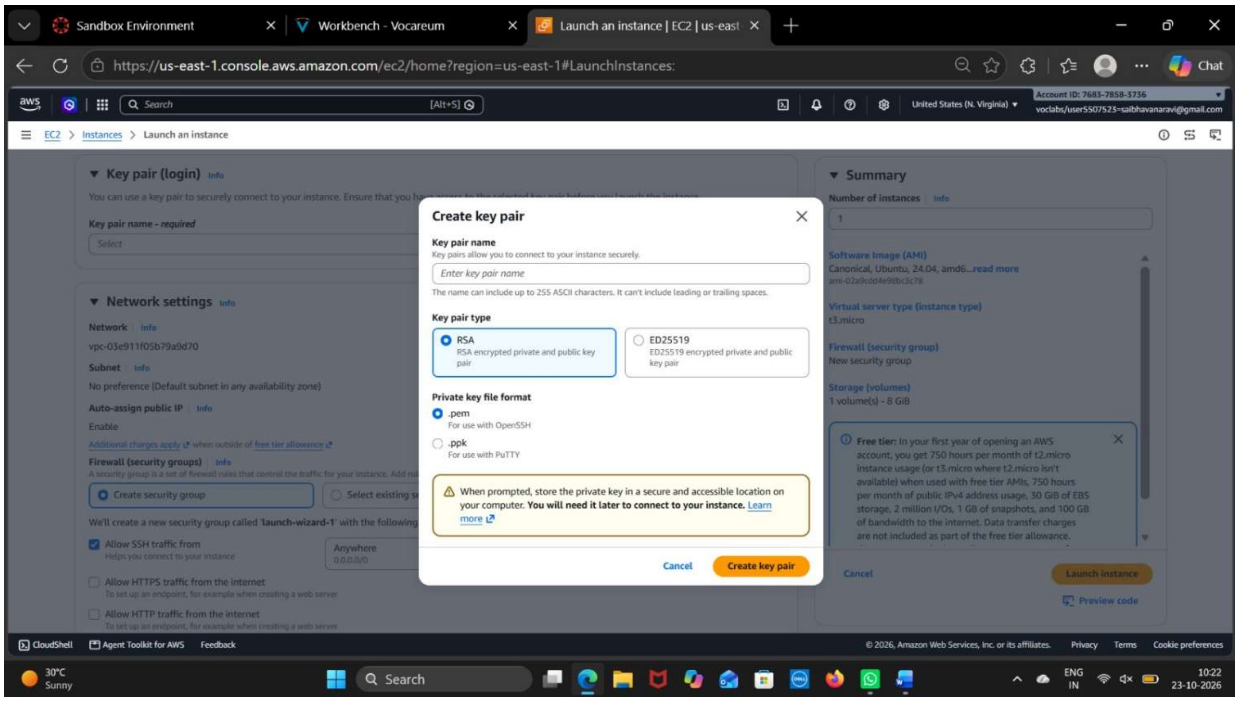


Step 6 — Launch or select an Ubuntu instance

If an instance has already been supplied, select it. Otherwise, choose **Launch instance**, select an Ubuntu Server AMI allowed by the lab, choose an instance type permitted by the sandbox, configure a key pair, and launch it. Follow the instructor's requirements for storage and networking.

Expected result: The instance appears in the Instances list and eventually reaches the Running state.

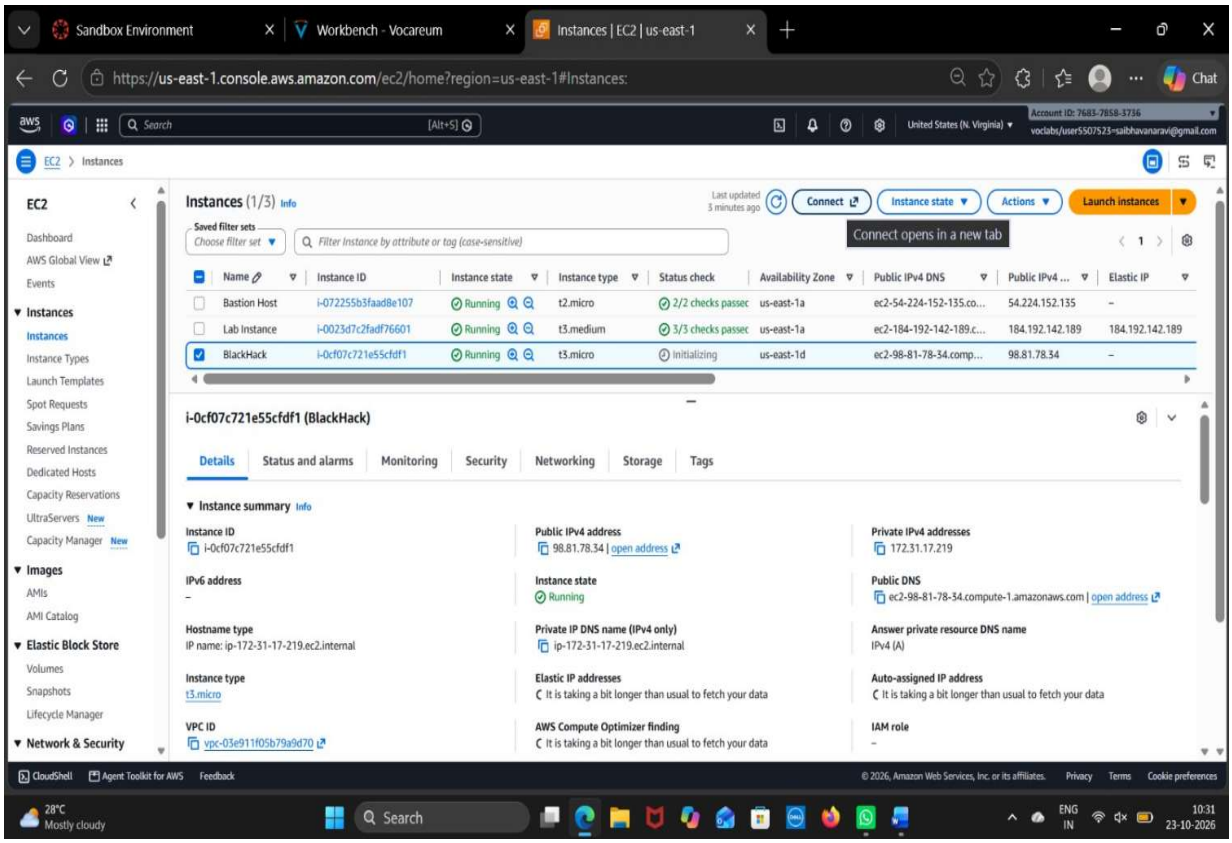




Step 7 — Record connection details

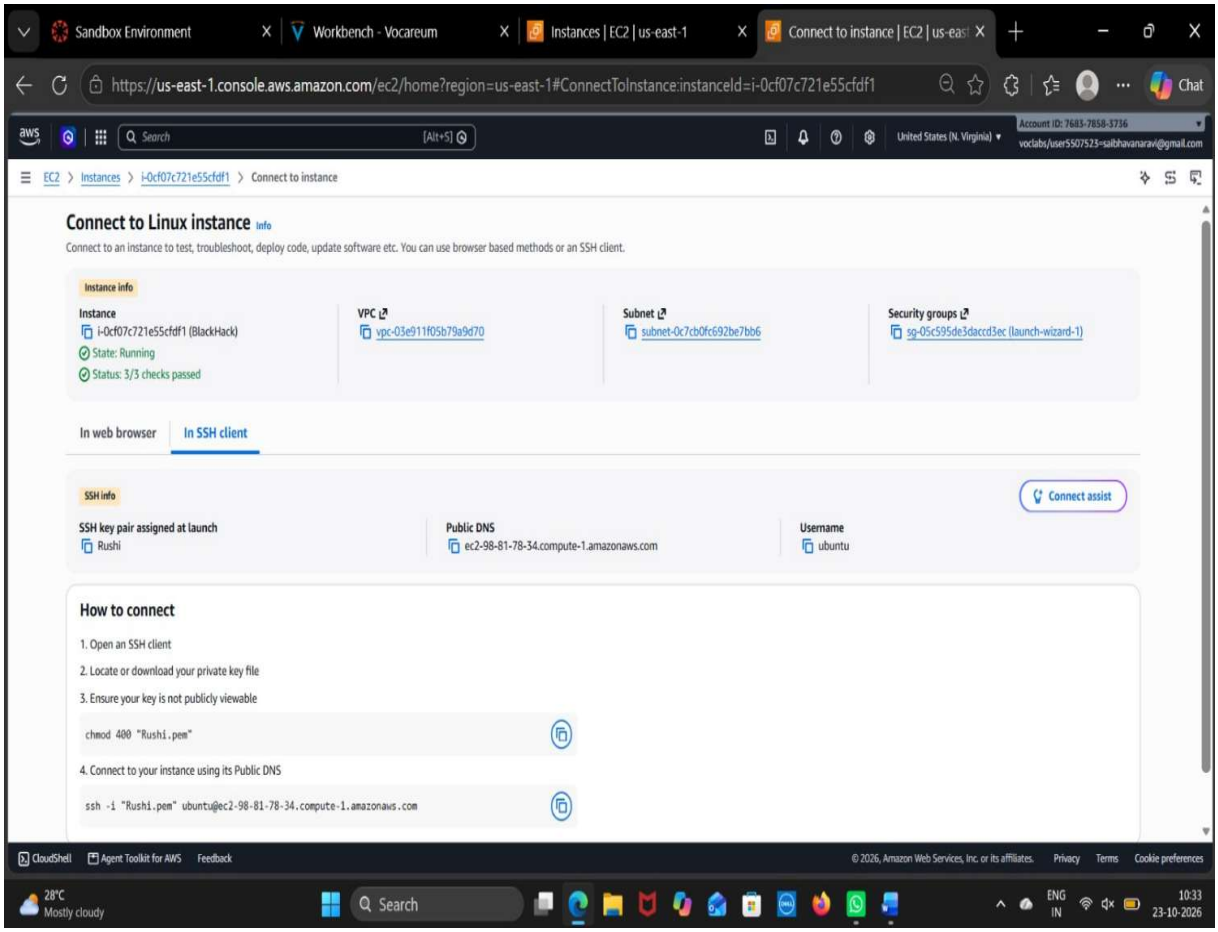
Select the instance and note its **Public IPv4 address** or public DNS name, its key pair name, and its security group. A public IP may change after stop/start unless an Elastic IP is used; Academy labs may restrict Elastic IPs.

Expected result: You have the correct public address and know which key pair belongs to the instance.



Step 8— Open the Connect to instance page

With the instance selected, click Connect . The page lists the instance, VPC, subnet and security group, the Public IPv4 address (98.81.78.34) and the default username **ubuntu.**



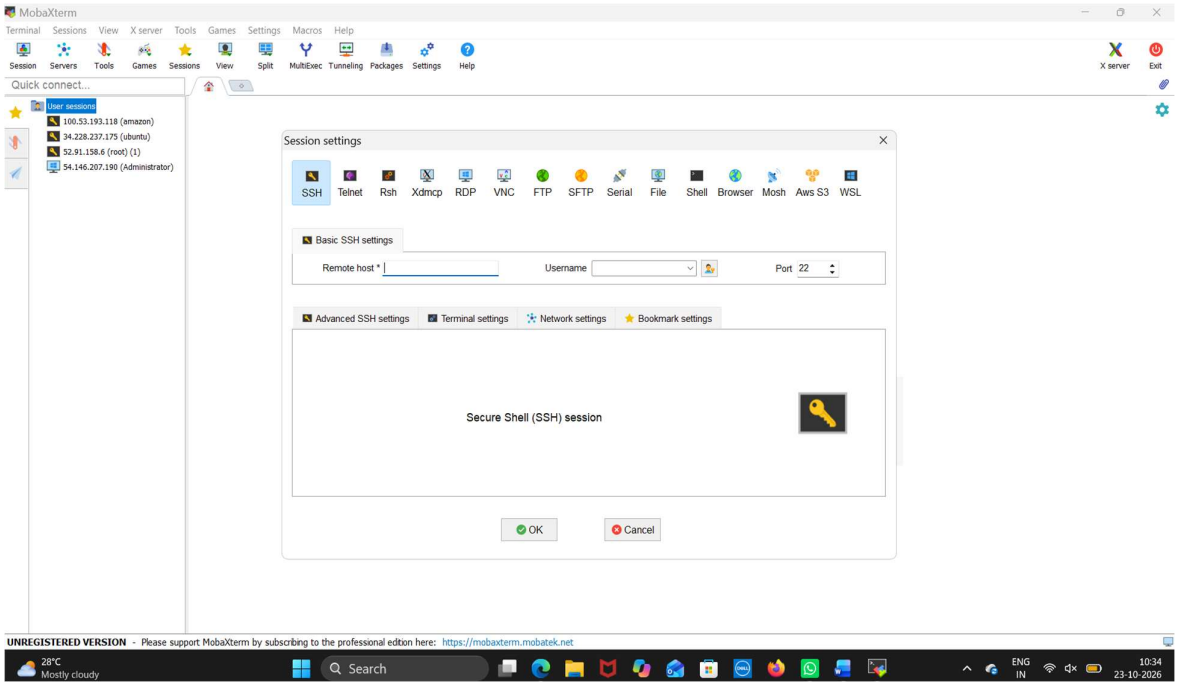
Part 3 — Connect using MobaXterm

Connect first with the normal Ubuntu account. Do not attempt root login until the initial connection works.

Step 9 — Create an SSH session

Open MobaXterm. Choose **Session** → **SSH**. Enter the EC2 public IPv4 address as the remote host and set the username to **ubuntu** for a standard Ubuntu EC2 image. If your lab image uses a different username, follow the lab instructions.

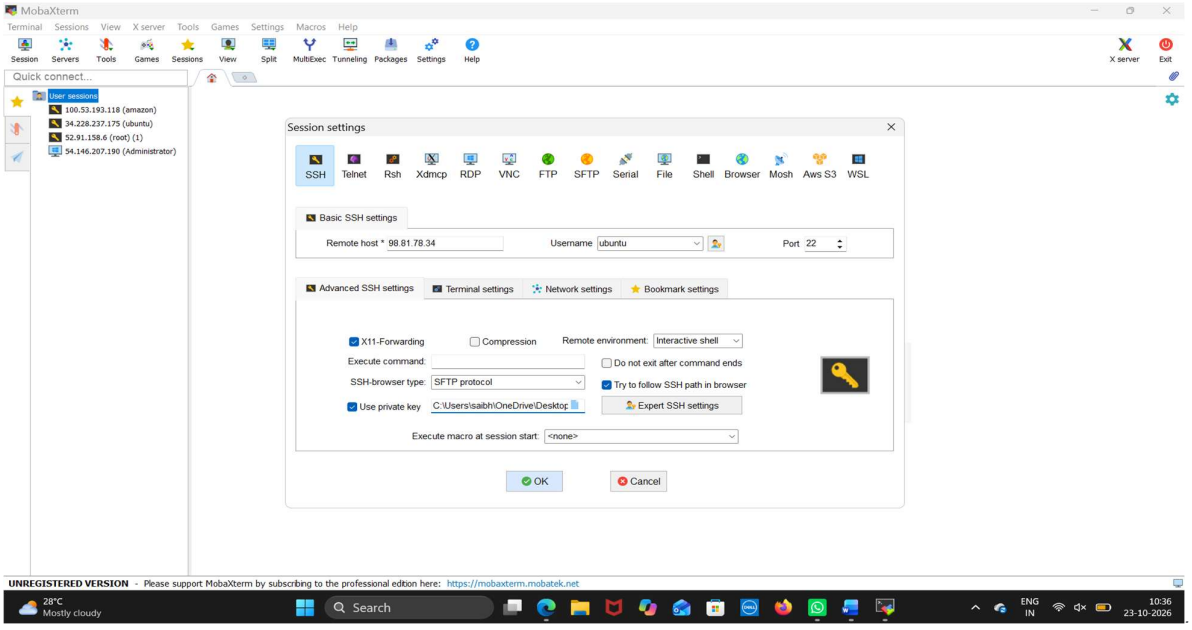
Expected result: The SSH session is configured with the correct host and username.



Step 10 — Configure the private key

Enable private-key authentication and browse to the private key file that matches the EC2 key pair. Some MobaXterm versions accept OpenSSH keys directly; if the key format is rejected, use the conversion/import option supported by your version. Never share or upload your private key.

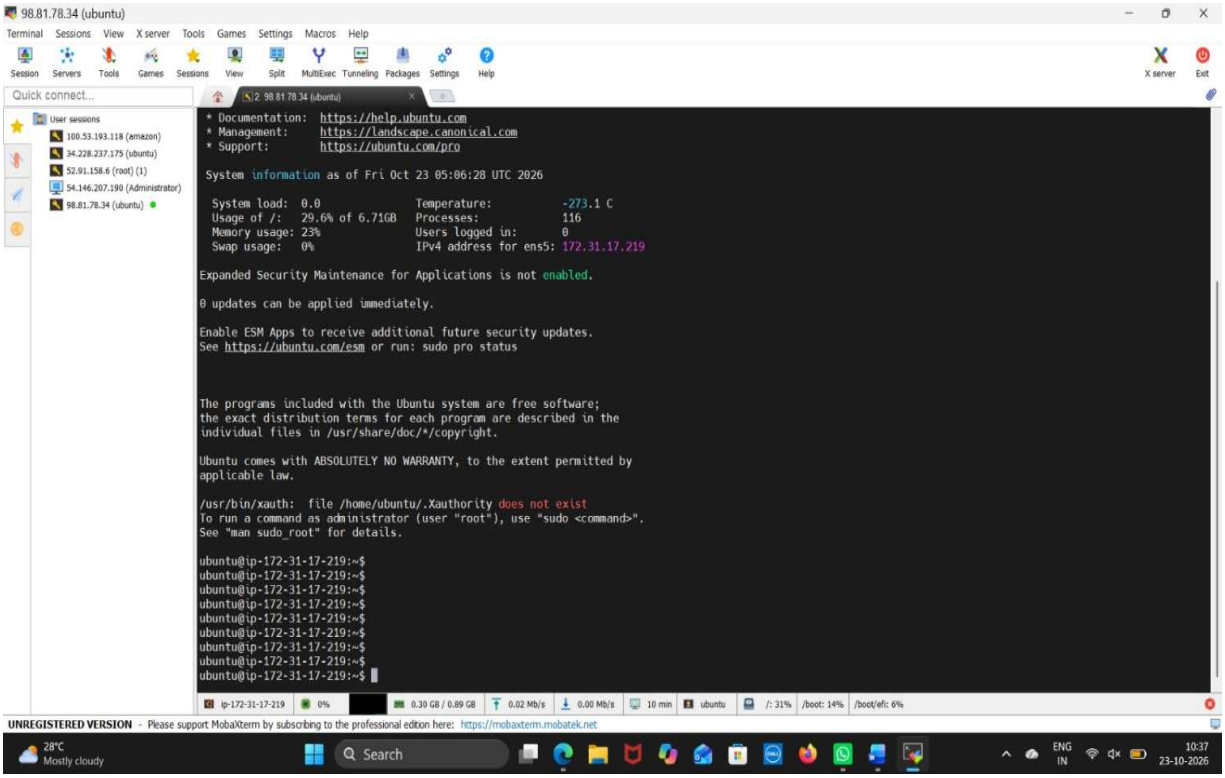
Expected result: The correct private key is selected for the session.



Step 11 — Connect and verify Ubuntu access

Start the session. On the first connection, check the host fingerprint prompt carefully and accept it only if you are confident that you are connecting to the intended instance. After login, u will see **$** which is a **privilege user** not root user.

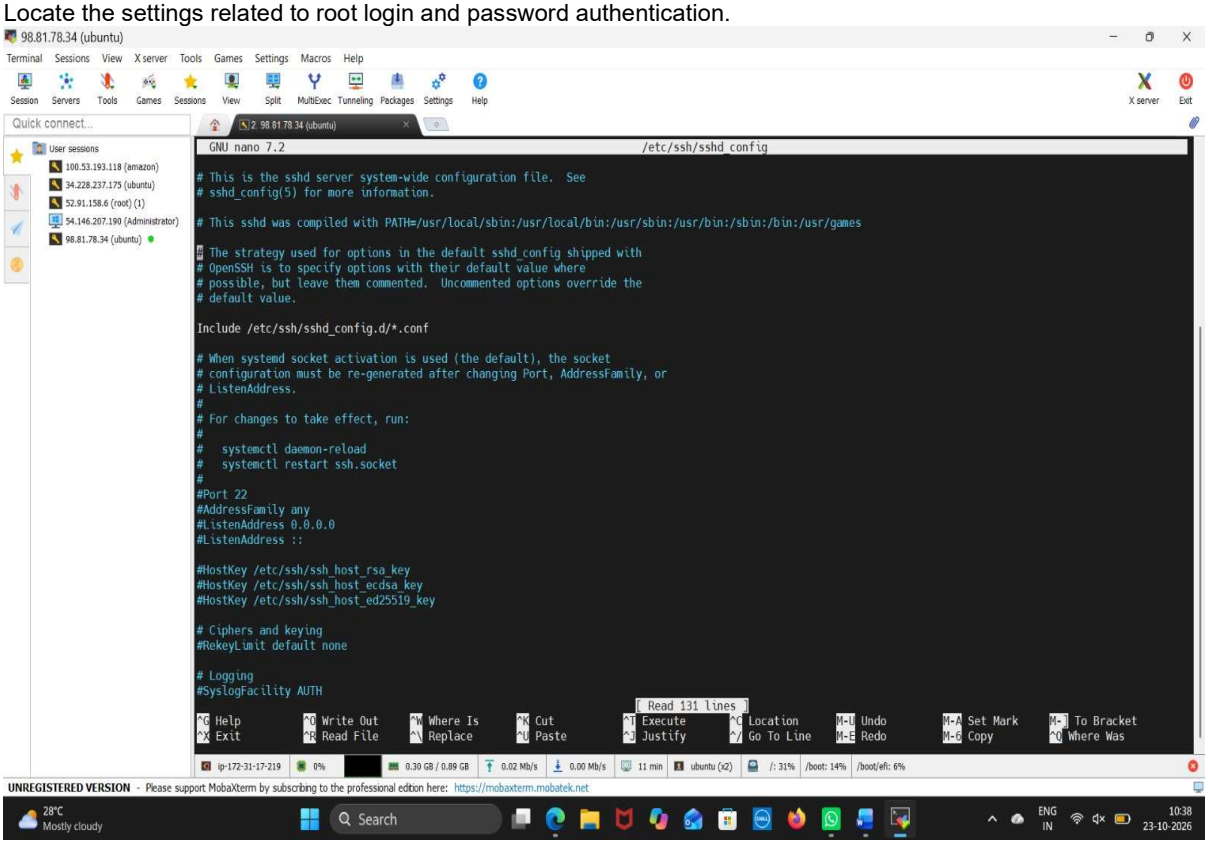
Expected result: The first command should show ubuntu for a standard Ubuntu EC2 image; hostname and id display the instance and account information.



From the Ubuntu session, open the SSH server Configuration with the command:

```
sudo nano /etc/ssh/sshd_config
```

Locate the settings related to root login and password authentication.



What this step demonstrates: **Purpose:** Configure the SSH daemon to permit the required root/password login.

Step 12 — Configure root and password authentication

In **/etc/ssh/sshd_config** , set the required options to allow direct root login and password authentication.

Remove the '**#**' before this four rules in the /etc/ssh/sshd_config.

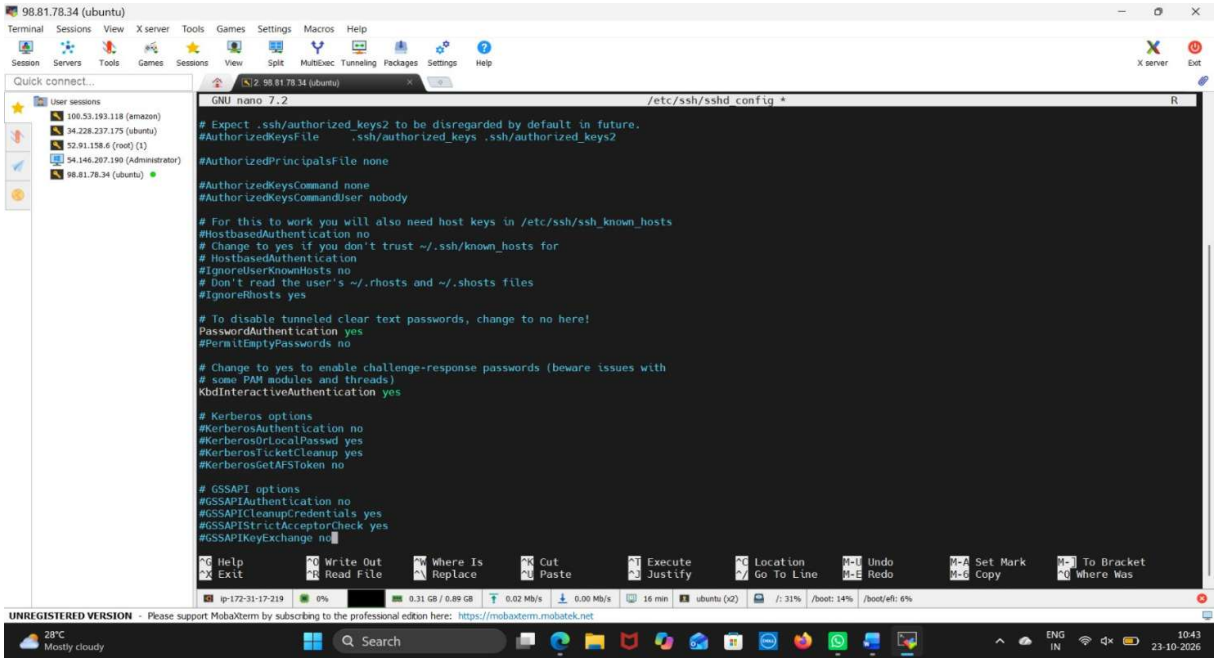
```
PermitRootLogin yes

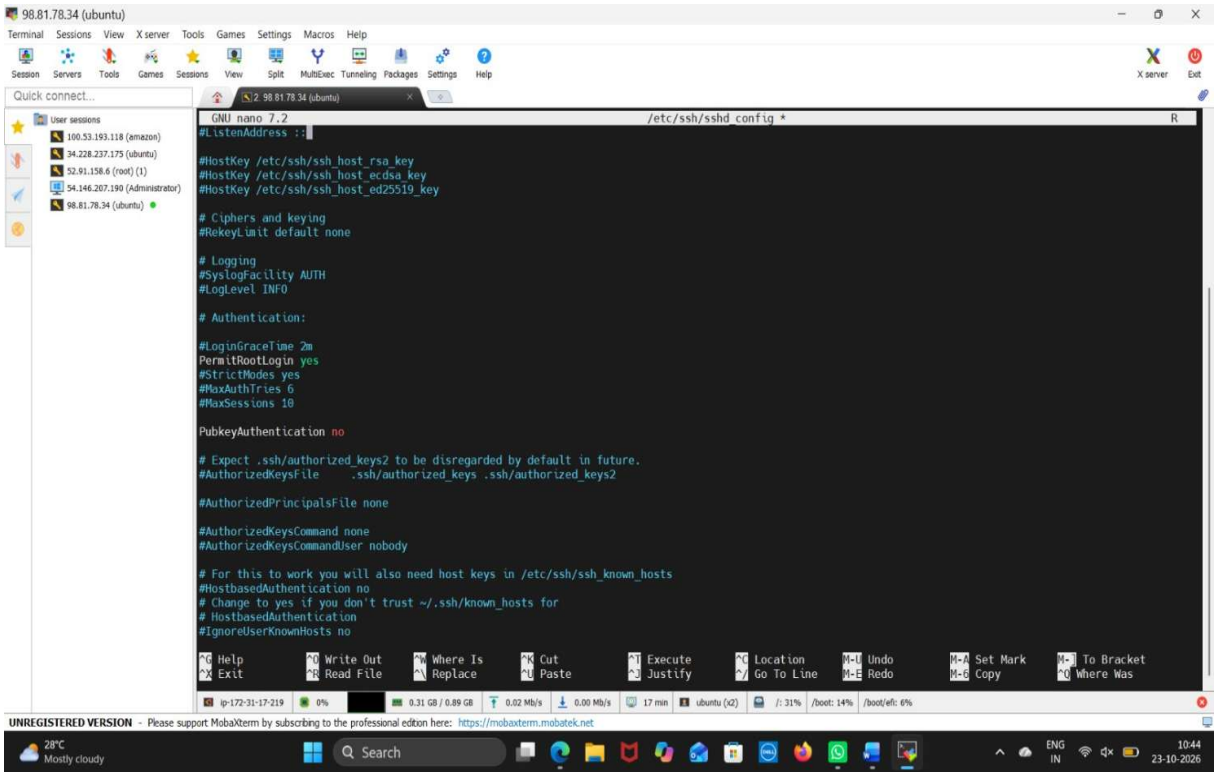
PubkeyAuthentication no

PasswordAuthentication yes

KbdInteractiveAuthentication yes
```

Save the file with Ctrl+O, press Enter, and exit with Ctrl+X.





Part 4 — Set a root account password

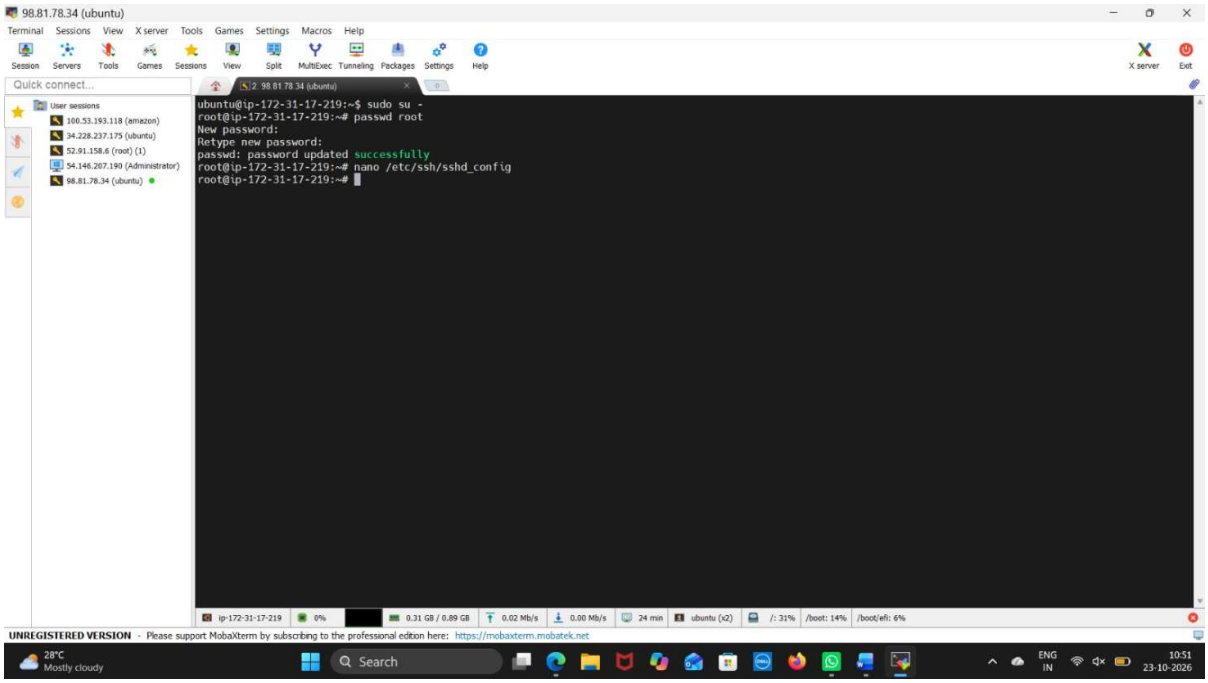
Use the authorized Ubuntu account with sudo privileges to set a password for the root account.

Step 13 — Set the root password

In the connected terminal, run the command below. Enter your Ubuntu user's sudo password if requested, then type a strong lab password twice when prompted. The characters will not appear while typing.

```
passwd root
```

Expected result: The command reports that the password was updated successfully if it completes without error. Do not include the actual password in screenshots or this report.



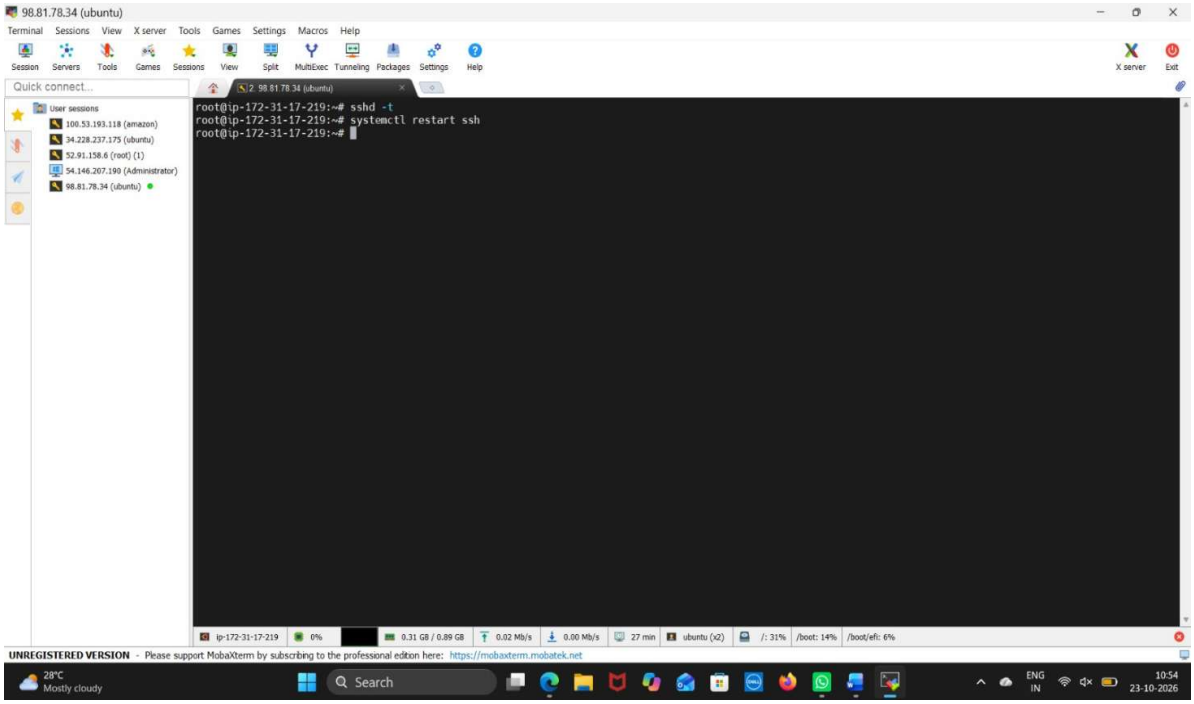
> **Security note:** Setting a root password does not by itself enable root SSH login. SSH server settings and authentication policy must also permit it. Do not reuse a personal password.

Step 14 — Validate SSH

Then validate the SSH configuration with sshd -t. If there is no output, the syntax check passed. Restart SSH With systemctl restart ssh.

```
sshd -t
```

```
Systemctl restart ssh
```



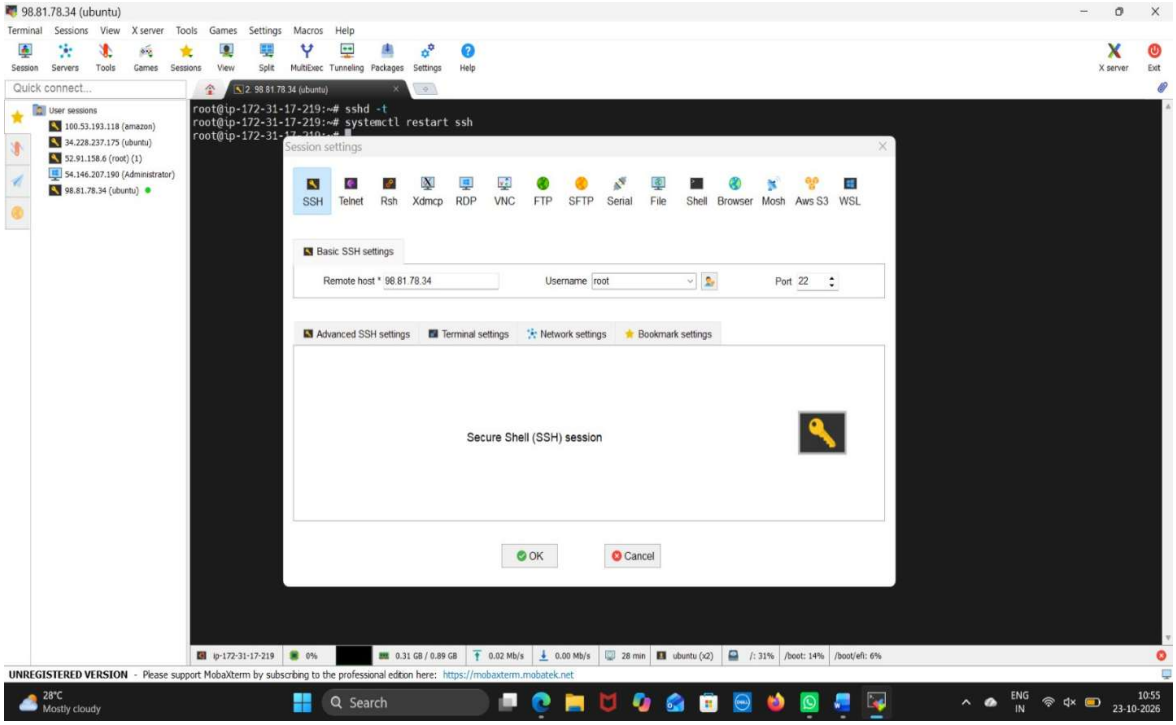
Part 5 — Test root SSH login and verify results

Use a second MobaXterm session to test the configuration while keeping the original Ubuntu session open.

Step 15 — Create a separate root SSH session

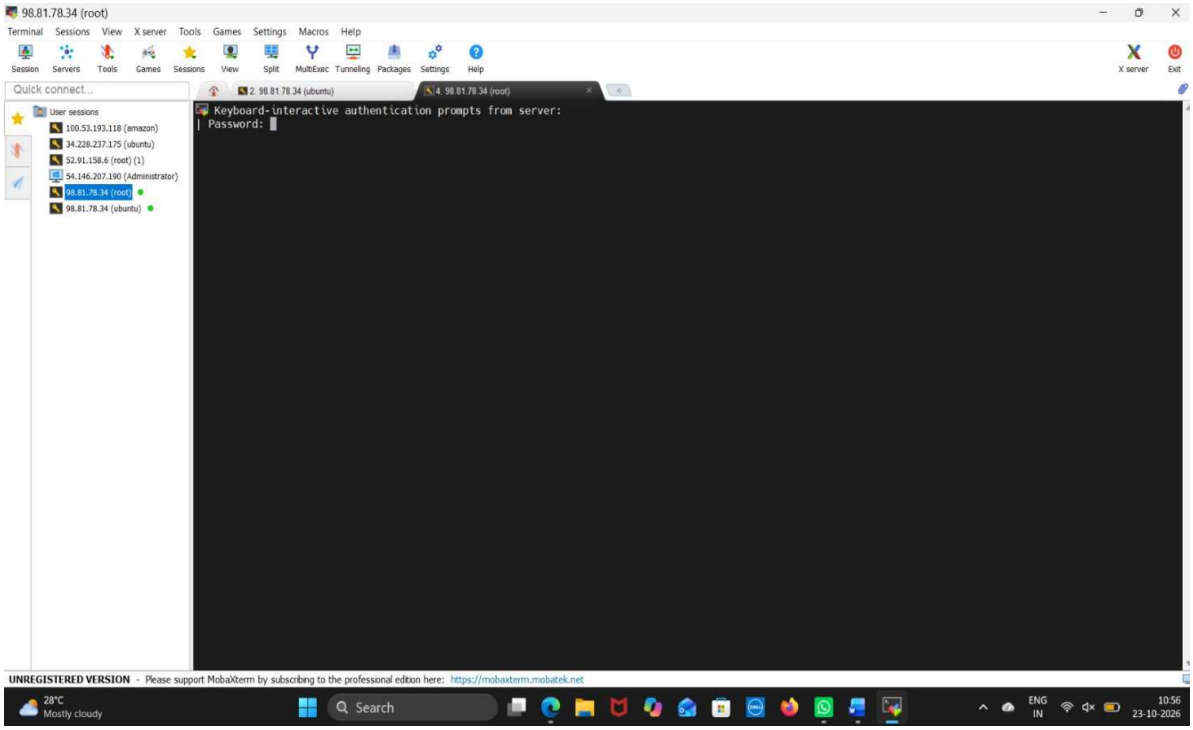
In MobaXterm, create another SSH session to the same public address. Set the username to **root**. For this test, do not select the Ubuntu private key as the only authentication method if you specifically need to test password authentication; allow the client to prompt for a password.

Expected result: The client prompts for a password or reports the server's authentication response.



Step 16 — Enter the root password

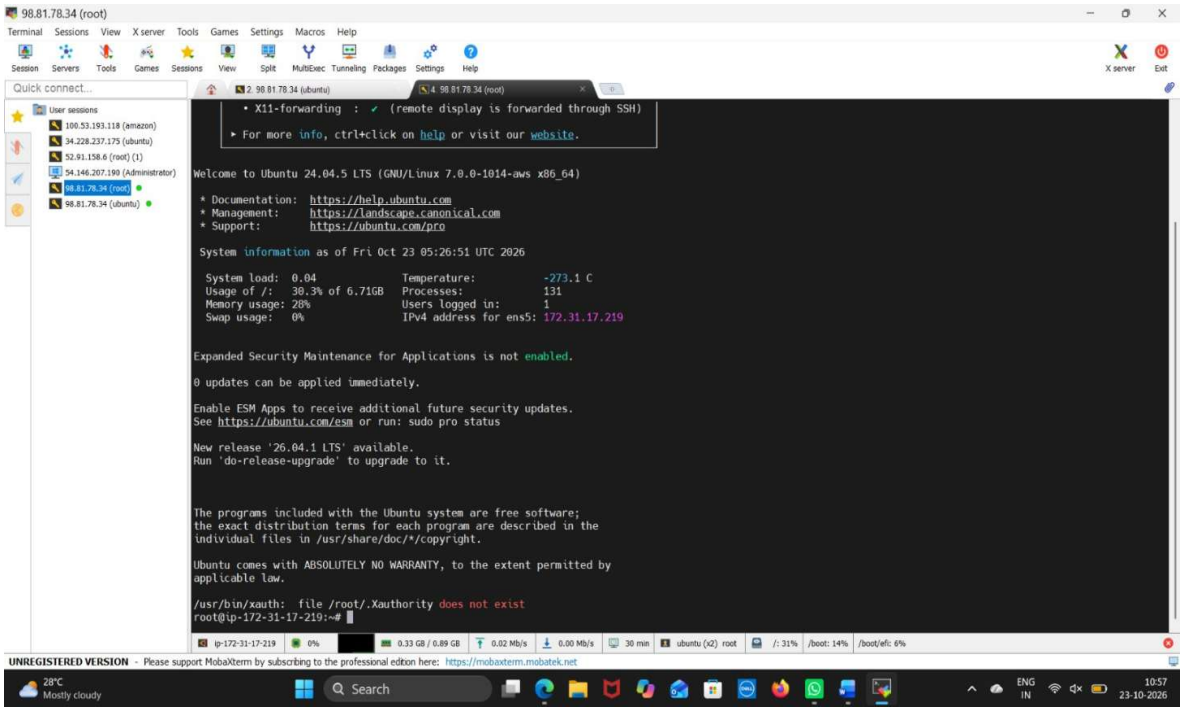
Enter the root password you set with `passwd root`. If login succeeds, verify the account with the commands below. Never place the actual password in the report.

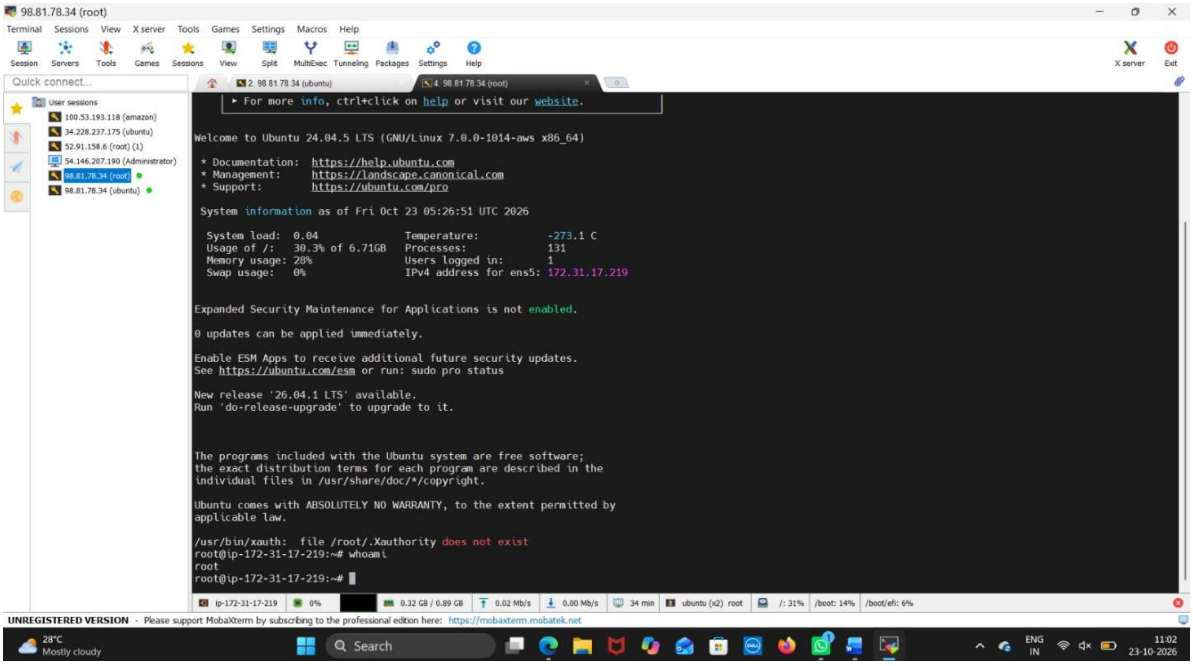


Step 17 — Final Verification Result

The final screenshot documents the completed direct root login. The server is accessible through MobaXterm as the root user without first logging in as ubuntu and running sudo su.

What this step demonstrates: Result: Direct root + password SSH login completed for the lab task.





Expected result: A successful root login returns root for *whoami* and shows UID 0 in the id output. If authentication fails, use the troubleshooting section.

Testing and verification checklist

- AWS Academy Sandbox started and AWS console opened.
- Ubuntu EC2 instance is running and its public address was checked.
- Security group permits SSH only from an approved source.
- Initial MobaXterm connection with the normal Ubuntu account succeeded.
- Root password was set using sudo passwd root.
- SSH configuration syntax was validated before restart.
- Root login test was recorded without exposing secrets.
- Safer SSH settings were restored and normal Ubuntu login was retested.

Conclusion

This practical demonstrates the relationship between Linux account passwords, SSH server policy, AWS networking, and client-side SSH configuration. The normal Ubuntu account with sudo and key-based SSH is the recommended administration method. Direct root password login is included here only as a temporary lab exercise and should be disabled after testing.

Security precautions

• Restrict port 22 to your own IP or an approved lab range.

• Never share private keys, root passwords, AWS credentials, or terminal screenshots containing secrets.

• Do not use direct root password login on a production server.

• Keep an existing session open while testing SSH changes.

• Restore the lab's original SSH policy before finishing and verify normal access.

References

- AWS EC2 User Guide: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/
- Ubuntu Server documentation: https://documentation.ubuntu.com/server/
- OpenSSH server manual (sshd): https://man.openbsd.org/sshd